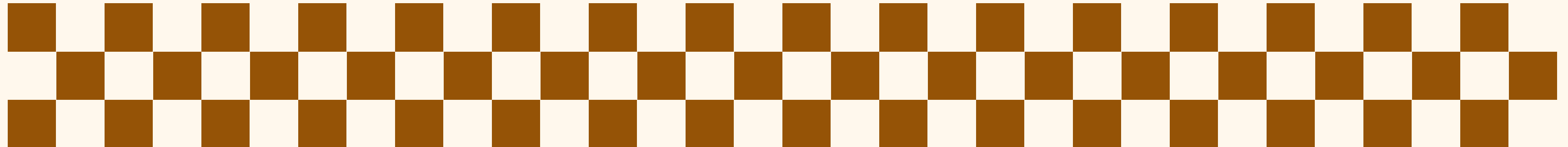


DETECTION OF PLASTIC LITTER ON BEACHES





PROBLEM BACKGROUND

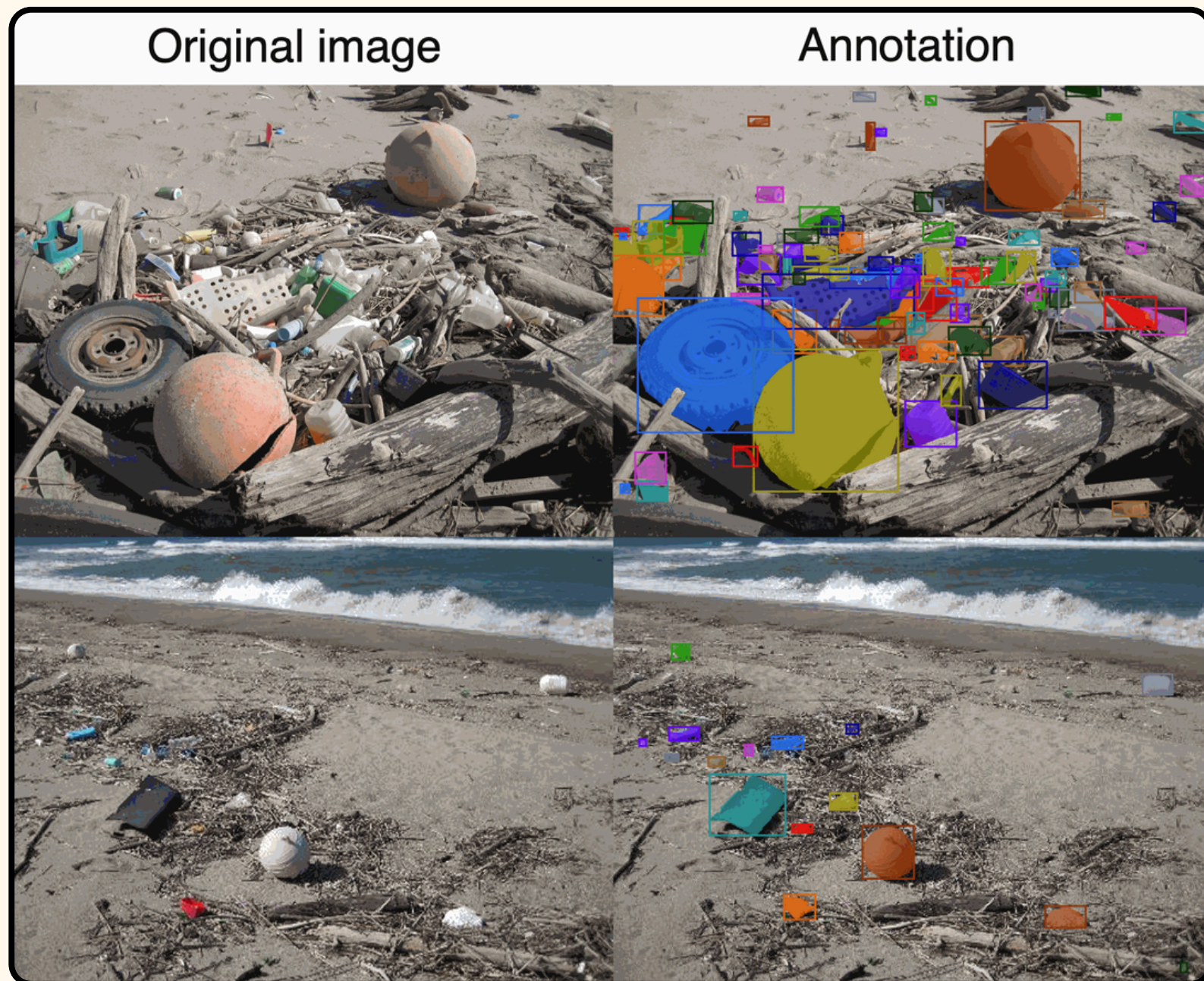


WHAT PROBLEM ARE WE SOLVING

- Plastic pollution threatens marine ecosystems
- Beach plastic is easily ingested by wildlife
- Manual beach surveys are slow and error-prone
- Automated visual detection is needed

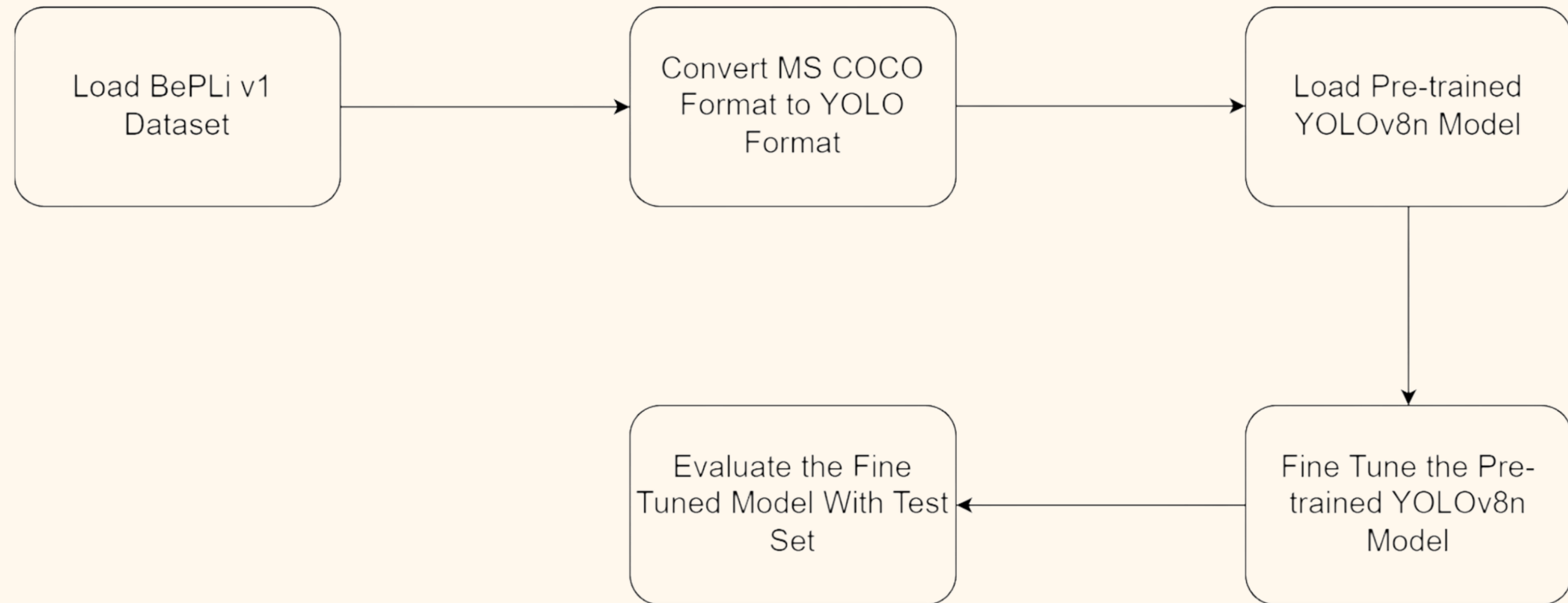
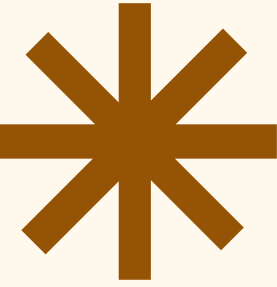
DATASET

Dataset: BePLi v1



- 3,709 real-world beach images (Japan)
- Single class: plastic litter
- Bounding box annotations
- Train / Validation / Test split

MODEL APPROACH



EVALUATION

Metric	Value
Precision	0.52
Recall	0.40
F1 Score	0.45
mAP@0.5	0.39
mAP@0.5-0.95	0.17



REFLECTION



Reflection & Limitation

- Model works on real-world beach images
- Struggles with precise localization
- Performance affected by object size and background complexity

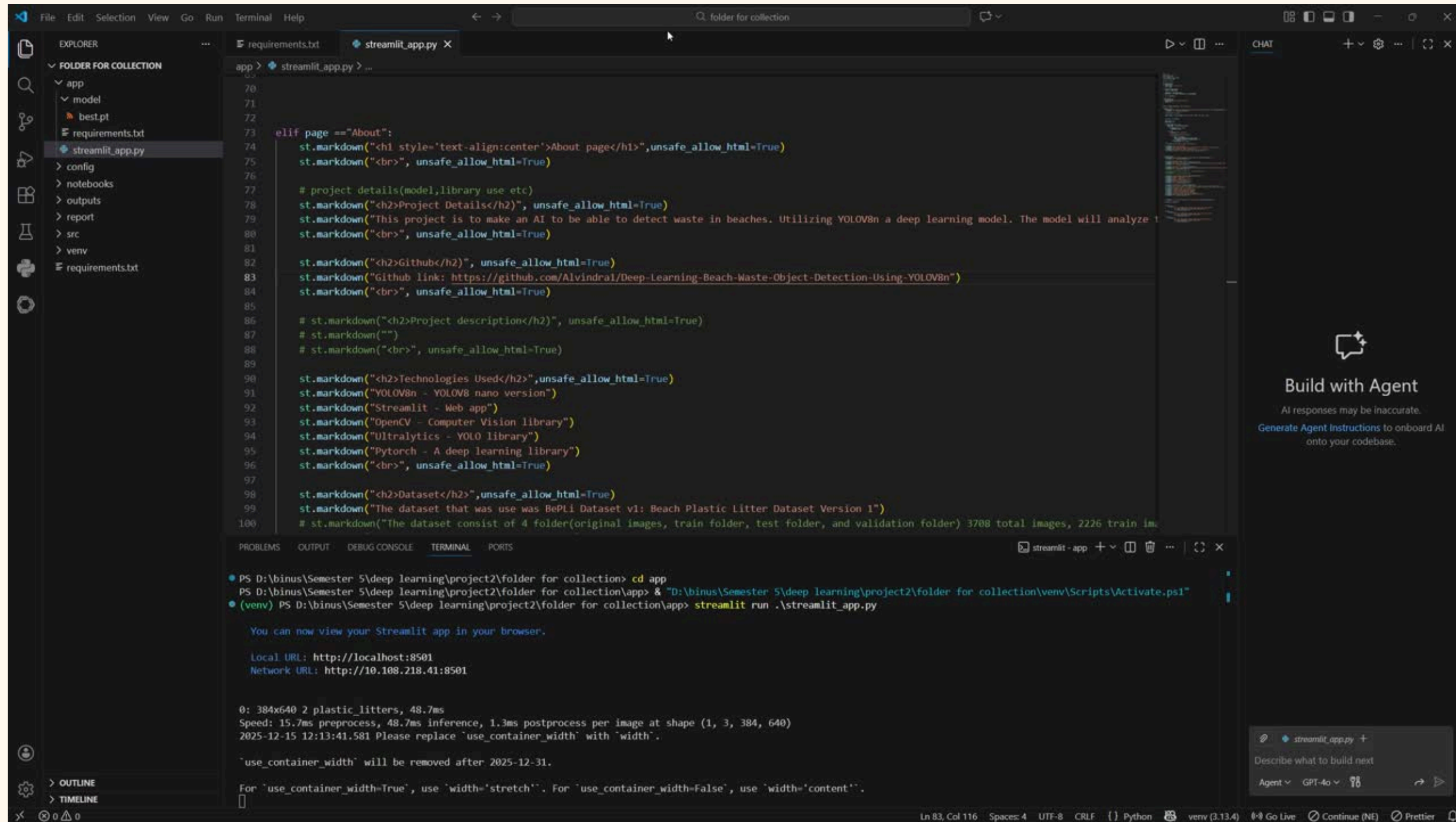
Future Improvement

- More diverse training data
- Larger YOLO variants
- Improved data augmentation

CONCLUSION

1. **Deep learning enables automated beach plastic detection**
2. **YOLOv8n provides a practical balance between speed and accuracy**
3. **Real-world environmental monitoring remains challenging**

DEMO VIDEO



The screenshot displays a Visual Studio Code editor with a Python project. The Explorer sidebar on the left shows a folder named 'FOLDER FOR COLLECTION' containing subfolders 'app' and 'model', and files 'best.pt', 'requirements.txt', and 'streamlit_app.py'. The main editor window shows the 'streamlit_app.py' file with the following code:

```
70
71
72
73 elif page == "About":
74     st.markdown("<h1 style='text-align:center'>About page</h1>", unsafe_allow_html=True)
75     st.markdown("<br>", unsafe_allow_html=True)
76
77     # project details(model,library use etc)
78     st.markdown("<h2>Project Details</h2>", unsafe_allow_html=True)
79     st.markdown("This project is to make an AI to be able to detect waste in beaches. Utilizing YOLOV8n a deep learning model. The model will analyze t
80     st.markdown("<br>", unsafe_allow_html=True)
81
82     st.markdown("<h2>Github</h2>", unsafe_allow_html=True)
83     st.markdown("Github link: 

The bottom panel shows the TERMINAL output:



```
PS D:\binus\Semester 5\deep learning\project2\folder for collection> cd app
PS D:\binus\Semester 5\deep learning\project2\folder for collection\app> & "D:\binus\Semester 5\deep learning\project2\folder for collection\venv\Scripts\Activate.ps1"
\(venv\) PS D:\binus\Semester 5\deep learning\project2\folder for collection\app> streamlit run .\streamlit_app.py

You can now view your Streamlit app in your browser.

Local URL: http://localhost:8501
Network URL: http://10.108.218.41:8501

0: 384x640 2 plastic_litters, 48.7ms
Speed: 15.7ms preprocess, 48.7ms inference, 1.3ms postprocess per image at shape \(1, 3, 384, 640\)
2025-12-15 12:13:41.581 Please replace 'use_container_width' with 'width'.

'use_container_width' will be removed after 2025-12-31.

For 'use_container_width=True', use 'width='stretch''. For 'use_container_width=False', use 'width='content''.
```



On the right side of the editor, there is a 'CHAT' panel with the heading 'Build with Agent' and the text 'AI responses may be inaccurate. Generate Agent Instructions to onboard AI onto your codebase.' Below this, there is a search bar and a list of files to chat with, including 'streamlit\_app.py'.


```

[Video Link](#)

THANK
YOU!

